



CSE461

Introduction to Robotics Lab

Lab No. : 02

Group : 06

Section : 04

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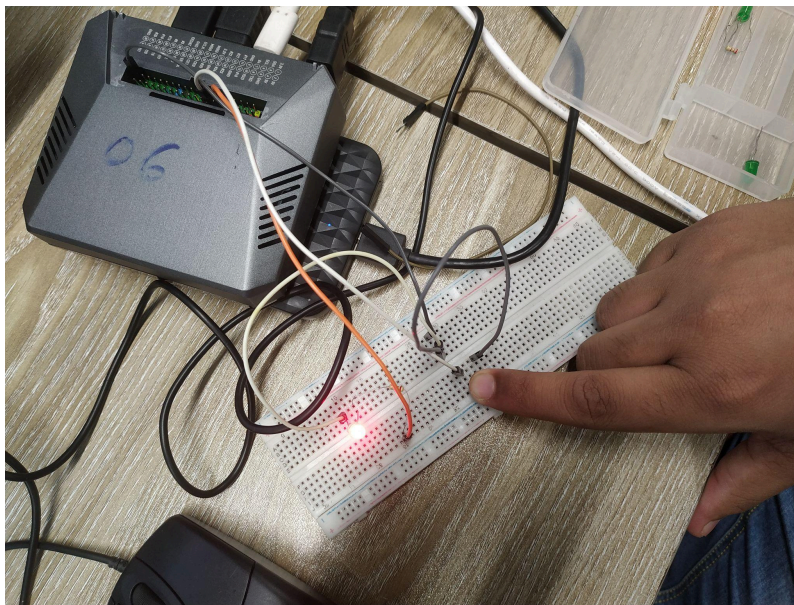
Submitted to - Aabrar Islam and Purba Tahia Hassan

1. Objectives: The aim of the current experiment is to create and construct a simple circuit using Raspberry Pi, which would switch an LED. When the switch is released, the LED will go off and when it is pressed, it will go on. The given task is intended to aid the comprehension of the fundamental application of the GPIO pins, input and output control and communication between the hardware and software components using Python.

2. Equipments:

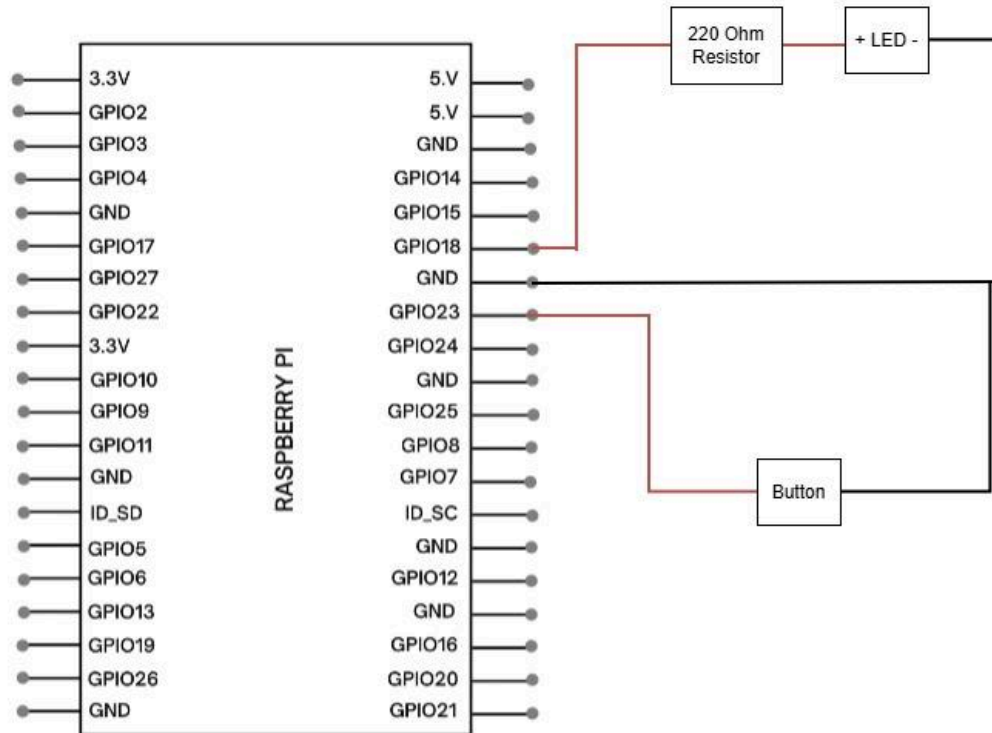
- Raspberry PI
- LED (Light Emitting Diode)
- Push-button switch
- Resistor (220 Ω for LED)
- Breadboard
- Jumper wires

3. Experimental Setup:



The LED is connected to the GPIO pin 18 together with a 220 ohm resistor, and the switch is attached to the GPIO pin 23. Pressing the switch will make the LED turn ON, and of course, releasing the switch will turn the LED OFF. The circuit is driven with the help of a python program that measures the input of the switch with the help of the GPIO pins.

Circuit Diagram:



4. Code:

```
# Enter Code Here
from gpiozero import LED, Button
import time

led = LED(18)
button = Button(23)

while True:
    if button.is_pressed:
        led.on()
        print("Switch pressed")
    else:
        led.off()
        print("Not pressed")
        time.sleep(0.5)
```

- 5. Results (Output of the experiment):** The experiment was completed successfully. The press of the push-button switch would change the state of the LED to ON and the release would change it to OFF. The python code was functional and displayed messages such as Switch pressed and Not pressed at the terminal. This proved that the Raspberry Pi was able to receive the button input and output the LED. The circuit was useful to illustrate the simple application of the GPIO pins to digital input and output.

Button State	Output	Terminal
Pressed	LED ON	Switch pressed
Released	LED OFF	Not pressed

6. Discussions/Answers:

This experiment served to learn how a Raspberry Pi can be used to turn on and off an LED with the help of a push-button switch with the help of the GPIO pins. The findings indicated that the LED was able to read the button press, and thus had elementary control of digital input and output. Certain difficulties were making sure that connections were correct on the breadboard and the correct use of the GPIO pins in the code. Solutions were found by making sure the circuit is well checked and that the pin numbers of the Python program were well checked. In general, the task was useful as it gave us hands-on experience with interaction between hardware and software, the lesson about the use of resistors to safeguard the components and better knowledge of Python-based programming with the use of the GPIO on real-world projects. The experiment supported the idea of observing inputs and regulating outputs in embedded systems in an effective way.